Chapter 18
Foreskin Restoration 1980–2008

R. Wayne Griffiths, J. David Bigelow, and James Loewen

Abstract The goal of foreskin restoration is to cover the glans penis to some extent with a double sheath of retractable tissue. Many men who contact NORM (the National Organization of Restoring Men) want full coverage of the glans, plus overhang, even when fully erect. A realistic goal is important, however, since satisfaction or disappointment is clearly related to expectation. For a tightly cut man, "success" may be just enough loose tissue so that erections are no longer painful. For others, it may be possible to achieve full coverage during erection. Currently, there are both surgical and non-surgical methods to re-cover the glans. The results, however, of most surgical procedures have proven disappointing. This presentation discusses both methods; however, the emphasis is upon various non-surgical tissue expansion techniques and devices. While the moveable sheath that covers the penile shaft is commonly called "skin," its structure is far more complex. Therefore, expansion of the shaft tissue is more challenging than expanding ordinary skin.

Keywords Foreskin restoration · Glans · Foreskin · Non-surgical tissue expansion · Moveable sheath · Dartos muscle · Peripenic muscle · Raphe · Corpus cavernosa · Corpus spongiosum · Tunica albuginea · Nerve supply · Ridged band · Meissner's corpuscles · Viagra · Surgical reconstruction · Scrotal tissue · Mitosis

Natural Structure

Before recounting the history and methods of foreskin restoration, I would like to discuss the structure of the shaft tissue and the effects of tissue expansion upon that structure. I will then describe the most common devices used for tissue expansion. Parenthetically, I will mention surgical reconstruction. I will also show some devices used as protection for the circumcised penis.

R.W. Griffiths (✉)
National Organization of Restoring Men, Concord, CA, USA
e-mail: Waynerobb@aol.com

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To understand the process and practical issues associated with restoration, we need to take a detailed look at the structure of the penile shaft tissue. The *corpus cavernosa* is bound by fascia, which determines the diameter of an erection. As we discuss various aspects of penile structure and foreskin, it is important to bear in mind that the shaft covering is a complicated structure, not just skin (derma) as most people seem to believe.

The dartos muscle starts at the perineum, envelops the scrotum, and, as it reaches the shaft, it is sometimes referred to as the peripenic muscle; albeit it is the same muscle. The dartos smooth muscle lies about 1 or 2 mm beneath the outer layers of the shaft tissue, which is skin. Between the dartos muscle and the skin are elastic fibers. The dartos muscle is quite separate from the skin layers, except on the ventral side where they join at the region of the median *raphé* (Fig. 1).[1]

The muscle fiber bundles are very slender and made of few cells, but they appear in every direction: transversely, longitudinally, and obliquely. At the prepuce, a great many of the fibers course in the long axis of the organ. They are loosely packed and filled with lax fibrous tissue. One researcher suggests that the dartos muscle fibers run circularly, forming a sheet of muscle of equal thickness for the entire length; however, the muscle lines the inner mucosal layer and doubles as the outer sheath tissue, forming the double layer of the foreskin.[2]

The layers of the penile shaft covering, from the outside inward, include the *epidermis*, *dermi*, and the superficial *fascia* (loose connective tissue that is connected to the *dermis* and enables the extraordinary mobility of the penile skin), which is also intimately connected to the *dartos fascia* layer (*peripenic muscle*), and to most superficial veins.

The next inward structure is the *fascia penis,* which is condensed connective tissue; it surrounds both the *corpora cavernosa* and *corpora spongiosum*. Each *corpus cavernosum* penis is surrounded by the *tunica albuginea,* very dense connective tissue and almost pure collagen, providing strength against over-inflation (in the manner of reinforced garden hose). This factor determines maximum penile size.[3]

The ridged band and frenular delta are the most specialized parts of the penis. They contain more nerve endings than any other part of the penile organ. There are

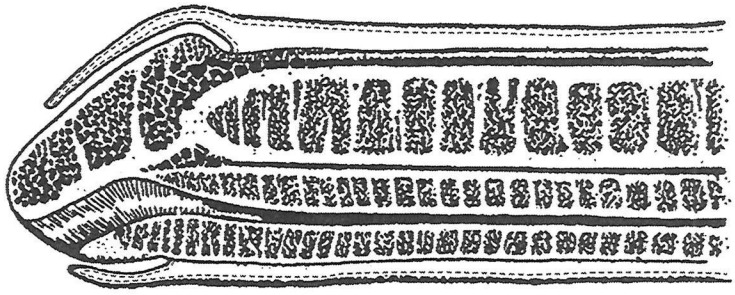


Fig. 1 Dartos muscle (see endnote 1)



Fig. 2 Ridged band (see endnote 5)



Fig. 3 Ridged band and frenular delta (John A. Erickson, photographer)

17.9 bundles/mm ventrally, 8.6 bundles/mm laterally, and 6.2 bundles/mm dorsally. As each bundle contains tens of axons, the nerve supply to the prepuce is obviously very substantial (Figs. 2 and 3).[4]

The majority of the medical community in the USA is apparently unaware of or disregards the fact that this specialized mucosal tissue even exists. They are

also apparently ignorant of the quantity of Meissner's corpuscles which, when stretched and rubbed over the corona of the glans, trigger fine touch sensations during sexual activity.[5,6] Circumcision removes this erogenous tissue, which cannot be replaced (although questions have been raised about the possibility of cloning a new prepuce). In my opinion, the medical community in the USA would not even consider this issue, and, if they did, it would no doubt be at the bottom of the list of their priorities. Also, the pro-circumcision advocates would not be helped in their cause. One restorer was told by his healthcare professional that the medical community prescribes Viagra-type medicines as an apology for circumcision. The medical community often uses terms, such as "redundant" and "excess" to describe the foreskin; however, a second set of adjectives that intact men know to truly define the foreskin include: "profuse, lavish, bountiful, luxuriant, exuberant."

Surgical Methods of Reconstruction

In order to fully cover the history of restoration, we need to at least mention surgical reconstruction—the chronology of which dates back to biblical times and also includes attempts during WWII to camouflage Jewish circumcision. Although a few modern-day procedures (beginning in approximately 1980) have been reasonably successful, most attempts have not had the desired results. A few types of reconstruction are (1) a skin graft from thigh or buttocks, (2) scrotal implant, or (3) Z- or Y-V plasties.

Skin graft methods place hairless tissue from elsewhere on the body, unlike penile/scrotal tissue, into a circumferential cut made around the penile shaft. Both the structure of the grafted tissue and muscle, if any, do not perform in the same manner as the very elastic penile shaft tissue.

The transplanted tissue usually has a very different condition, texture, and is quite inflexible and smooth. One man commented that his reconstruction looks great, but has no feeling and is quite numb. He added that his sexual activity has not been enhanced.

A scrotal implant graft is a multiple-stage reconstruction, involving circumferentially cutting the shaft tissue at the circumcision scar. A tunnel is created in the front side of the scrotum between two incisions, and the penile shaft is threaded through the tunnel, and stitched at both ends.

After about six months, when healed, the penis is surgically removed with the new scrotal tissue cut on either side and wrapped around the shaft and sewn on the ventral side. There is then another healing period. At that point, it is typically necessary to reduce the "overhang" and to enlarge the orifice of the new foreskin. One restorer, who had this procedure done in the mid 1980s, had a "wonderfully" long foreskin; however, it was non-retractable. Later, the plastic surgeon cut a ventral slit to facilitate retraction, leaving a dorsal flap over his glans of considerable length (approximately three inches).

Non-surgical Methods of Restoration

Contrary to surgical methods, tissue expansion uses the current tissue expansion method of producing new tissue. In the case of foreskin restoration, it avoids surgery, nerve loss, or disasters that are more common with surgical methods. The works by Takei et al.,[7,8] indicate that cyclical periods of tension are quite successful and a preferred method of producing additional new tissue. This method is well suited to the many and varied functions of the penis.

Parenthetically, some men have felt that penile pumping would achieve the desired results of tissue expansion; however, the new cells that form after pumping remain in a bloated state and continue to reproduce bloated cells for life. A couple of men who have pumped for many years have a penis that looks like a bloated sausage. In addition, the penis is soft and spongy to the touch.

A Logical and Successful Restoration Regimen

New cells are produced by the process of *mitosis*, that is, cell division. It has also been shown that excessive tension does more damage than moderate tension. Excessive tension causes scar tissue to form, which takes longer to "heal" and hinders flexibility.[9]

Since the shaft tissue is both muscle and skin, the model for foreskin restoration is a body-building regimen. In such a regimen, one works on the upper body one day and the lower body the next; this allows each set of muscles to have a day of rest to generate new cells and then to be coaxed again the following day to grow more cells. The same principle is true for the penile shaft tissue, which consists of tensioning for a determined number of hours during the day and then letting it rest at night.

Consequently, one must realize that there is *no* fast or instant restoration. One must work into his daily habits a regimen that will suit his lifestyle and work habits. One colleague has noted that one is "fooling" the penile tissue into "thinking that it has to grow to cover the longest penis in the world."

A suggested regimen (which has worked quite successfully for many restorers) is to apply weights or an elastic tether in the process of getting ready to start one's day. One wears them from four to eight hours and then removes them later in the afternoon or evening. Actually, it is suggested that one start his restoration regimen with a manual tensioning method to get accustomed to a daily regimen. By so doing, one forms a daily habit of tensioning. During this initial period, one has time to consider the type of device that would suit his lifestyle best before making a purchase or designing a homemade device.

It is important that cautions be observed and followed: Don't cause or endure pain. Don't be overzealous. Don't cause constriction of blood flow, which includes pain and/or color changes. It is important to make pressure tests to check blood flow.[10] Further, as part of the restoration regimen, no matter what device is being

used, there must be frequent release of the tension against the sheath tissue and/or pressure against the glans so as to allow for free blood circulation.

The restorer will want to keep the glans covered as soon as it becomes more sensitive and the nerves become more responsive. When one achieves some longer tissue, a retainer can be used for the remainder of the evening and night. This will keep the glans covered and protected from rubbing against clothing. It will also help retain the sub-preputial fluids, where they will start the process of dekeratinization (sloughing off of the layers of calloused tissue). The retainer might be a tape ring, crisscross taping, an O ring, or narrow Velcro strap. Such a nighttime rest period will allow the shaft tissue (*derma* and muscle) to produce additional cells.

Some restorers have advocated excessive tension for 24 hours, 7 days a week. This type of regimen does not work well, as most of us have other things to do, which involve our genitals, such as urination, sex, bathing, and rest. As noted above, damage and much longer healing time is required with such a stringent regimen, and constant tension is not necessary to achieve restoration. Experience has also shown that restoration does not happen faster with such a regimen. Some of the tissue and muscle may be torn by such excessive tension and, if so, scar tissue is formed in the muscle fibers; flexibility and mobility are thus reduced.

Every male has just one penis and needs to become aware of its structure and its care. Please be careful and be aware of what your own body is telling you about your efforts to restore.

There are many devices that have been invented since NORM was founded in 1989. In past centuries, the Pondus Judeaus and thongs were used to lengthen the foreskin or to keep the glans covered. The first modern-day recorded attempt occurred in the early 1980s. It involved implanting a small platinum ring in the remaining shaft tissue, which held the tissue in front of the glans. It was hoped that, eventually, the shaft tissue would lengthen to cover the glans when the ring was removed. The tension proved to be insufficient to lengthen the tissue, and, further, calcification formed a ridge where the ring was implanted.

A great many devices have been designed since that early effort. A select few of the current devices have been chosen for discussion, along with an evaluation of the results that each one produces. Various devices produce somewhat different results; all devices will produce new tissue and, thereby, produce a "foreskin." Many current procedures require the application of tape to the shaft tissue; however, fixed tape does not allow the tissue covered by the tape any tension or growth. Therefore, the less tissue covered by tape or the device, the more sheath tissue is tensioned and coaxed to grow additional cells. As a consequence, the degree to which tape is used in the various procedures should be taken into consideration when choosing a restoration method.

Restoration Devices

The following is a brief description of the selected devices in the approximate order in which they were invented and made available to the public. Only a few devices have actually been patented: the PUD, Vac-U-Trac, and the Tug Ahoy. Most others

are being produced and made available without restriction. At present, NORM is aware of at least 17 devices available, and a more thorough count, no doubt, would yield several more.

The FOREBALLS was devised in 1987. The first version was made by "barrel" taping two balls together as is shown in *The Joy of Uncircumcising,* by Jim Bigelow, PhD, as a homemade device. With ideas from fellow restorers at NORM meetings in San Francisco in the early 1990s, the tape was replaced by a rod welded between the two balls. At one of the meetings, someone suggested the name "FOREBALLS."

The PUD (Penile Uncircumcising Device) was patented by American Body Crafters' Roland Clark. The device was made in several sizes/weights. Wide tape is required to secure the device to the shaft tissue. Because of the circumference of the device, it holds the new foreskin in a somewhat tube shape. The American Body Crafter web site touts it as, "The most sophisticated foreskin restoration product to date."

The BUFF (Brothers United for Future Foreskins) method was available privately from a source in Arizona. It featured the use of tape strips to secure the foreskin in front of the glans. It was soon discovered that a source of additional tension was needed to produce further results. The use of silicone cones to provide the needed tension was one of the first methods developed. It was published in Mark Waring's 1988 pamphlet, *Foreskin Restoration (uncircumcision).*

After several months of utilizing the information from BUFF and Waring's pamphlet, Tim Hammond and R. Wayne Griffiths felt the need for a support group. Together, we founded NORM in February 1990. Shortly after its founding, the NORM group was asked to try on various sizes and shapes of the Second Skin Cones, which we did at a meeting in Concord, CA. The cones were a bit bulky, and we made several recommendations for better shapes and sizes. When the cones were produced, in both weighted and non-weighted forms, men who used them complained that the meatus was pushed into the urinary opening and that long-term use was often uncomfortable.

D. Evans, of Arizona, devised the T-Tape System. The T-tape method uses fixed tape to secure the shaft tissue and provides tension with an elastic strap. In this method, the tape is applied around the shaft tissue of an erect penis (to allow for nocturnal erections) and holds the tissue in both directions, extended circumferential as well as longitudinal. D. Evans also made a small number of videos, not only of the device, but also of a NORM meeting, as well as other documents about the benefits of restoration. However, he soon went out of business, and others have since made available the instructions and T-tapes.

The RECAP-EZ was devised by Arthur Gibson of Texas. It is a combination device that allows the restorer to vary his regimen. The variations are all based on the cone-and-cup structure, which is a tapeless device system. One of the varieties has a scrotal retainer to hold the scrotum in place against the body and allow the cone/cup to tension only the shaft tissue. Due to their bulkiness, most of the RECAP-EZ devices need to be worn at home.

The DTR, Dual Tension Restorer, uses pressure against the glans to provide tension; however, the strength and size of the rubber band determines the exact tension. NORM holds that using pressure against the glans to achieve tension is not the best

procedure. The glans is being crushed, thereby, compressing nerve endings and cutting off circulation. One must be sure to relieve such pressure/tension often in order to provide relief from the compression and allow for blood circulation.[11]

The CATIIQ—(Constant Applied Tension II Quick), devised in 2002, is another variation on the pressure-against-the-glans system. It uses a rod with a collet/ferrule that applies whatever tension one puts on the glans.

TUG AHOY is made by a physician in southern California, which he touts as the "gold standard of foreskin restoration." The cup is made from half of a flexible toy-store "egg." The other half is used as the cone, with a bent coated wire attached to an elastic strap for tensioning.

The TLC TUGGER is a very sophisticated tapeless tensioning device made with various sizes of cups and cones to fit the individual shape and size of the glans penis. It is made of food grade silicone and uses an elastic band for tension.

THE NATURAL RESTORER is a weighted device of stainless steel with a long rod. A lock screw on the rod governs tension. It is a device that, again, puts pressure on the glans for most of the tension.

MYSKINCLAMP is a complex device. The cup puts pressure on the glans and the cone holds the sheath tissue. There is a sturdy spring that has to be adjusted and locked in place with a setscrew on the rod. Should the setscrew loosen, however, the spring would expand to the fullest, and that could possibly cause tearing and pain.

Homemade Devices

There are many inventive homemade devices made from the use of pill tubes, 35 mm canisters, and plastic pipe couplers, which are discussed further.

One device is made with a baby bottle nipple fitted over a cup with a rod through it to hold a ring. It uses an elastic strap for tension. It has the advantage of pulling away from the glans.

Paul's Foreskin Restoration Kit supplies materials and instructions to make a cup-and-cone tensioning device.

Tube devices are made using a pill tube, a PVC pipe coupler, a 35 mm canister, etc. to place over the glans. The sheath tissue is taped to the outside of the tube. It does not have to be removed for urination. While the device does lengthen the new tissue; it also widens it. Therefore, the lengthening tissue will not readily close down over the glans like a normal foreskin. Men have contacted me to get information on surgical closure because they have used such devices/methods that have not given closure to the aperture of the new foreskin. Although such surgical procedures are presented in *The Joy of Uncircumcising,* later experience with such surgical touch-ups has not proven successful. They are no longer recommended.

O rings have been used in various ways as nighttime retainers or several in a sequence on the foreskin to put tension on the tissue. If the rings are narrow and too small, circulation will be cut off quite quickly and the rings must be removed.

ANGUS' O RINGS are a variation on the use of O rings with a urination tube.

Foam cones are made in various lengths and widths to be compressed over the glans. The shaft tissue is then taped in place so that the foam expands to put tension on the shaft tissue.[12] Such cones, however, expand the tissue in both diameter and length.

Tape straps with weights or with an elastic band are made in any number of variations.

Single Ball Bearings. Some men have used a single stainless steel ball placed in front of the glans, and the shaft tissue is pulled over it. A Velcro strap or cross tape system is used to hold the bearing in place as a weight to tension the shaft tissue.

Velcro straps or tape rings are usually used as a retainer to keep the glans covered when not tensioning. It is similar to the O rings used for the same purpose.

Non-restoring Covers

There are a number of products that have been offered since the beginning of this era of restoration, such as:

MANHOOD is a rayon sheath that was designed to cover the circumcised penis and to protect it from abrasion and cold especially for bike riders. It was designed by and is available from the maker in Canada.

SENSLIP is a latex tube with ridges and is attached to the penis with a fold in it to simulate a functioning foreskin. It is somewhat complicated to apply. The concern is that the glans, mucosal tissue, and shaft tissue are covered with a foreign product rather than natural tissue. The manufacturer says that it is porous and should be used mainly for sexual intercourse. The problem, however, is that, if left in place too long, it produces an anaerobic condition that is not conducive to healthy tissue.

Natural Phimosis Treatment

NORM has been the central clearinghouse for information about both restoration and how intact men can lengthen their foreskin, as well as providing help for men with phimosis. For the latter, the following devices are available:

GLANSIE, from Japan, is a speculum (an instrument for examination of canals) with narrow tips that can be inserted into a phimotic foreskin. It puts tension in a lateral direction to gently open the foreskin so that it will allow retraction over the glans. With time, new cells are formed and the opening widens.

PLATIGO AFS BALLOON, from Australia, is a device using small balloons to put tension on the foreskin to widen the opening. With time, new cells grow in the foreskin tissue.

This homemade phimosis treatment device is merely an old-fashioned spring clothespin that has had the finger ends tapered. When used, it is opened and coins or washers are inserted into the opening. As necessary, these coins/washers are removed to open the tapered end to provide tension on the opening of the foreskin.

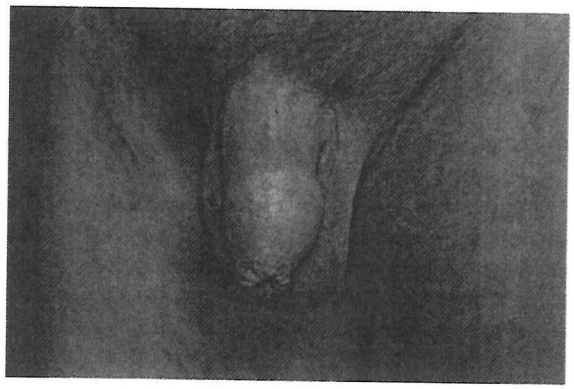


Fig. 4 Restored foreskin (James Loewen, photographer)

Results

As can been seen, the restorer in Fig. 4, after using tissue expansion for about a year and a half, has full coverage of the glans, with an overhang of about three quarters of an inch. When the body is warm, the foreskin opens and can be easily retracted.

It is also noted that, after 20 years of being covered, the foreskin has reduced in its thickness and works quite naturally. It protects the glans.

Notes

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